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Lower Limb Orthoses

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SUMMARY PEARLS

- Lower extremity orthoses provide biomechanical support for patients' legs and lower trunk, allowing them to stand and ambulate or to prevent further joint damage.
- Ground reaction forces are utilized from patients' shoes to their hip joints.
- Some lower extremity orthoses are utilized to prevent injury or further joint damage.
- Technology allows patients with spinal cord injuries and hemiplegia to stand and ambulate.
- Assistive devices are an essential part of allowing patients to gain independence safely.
- Pediatric patients are not small adults; they have special considerations for rapid changes and require more frequent follow-up by the rehabilitation team.

Introduction

Lower limb orthoses serve multiple purposes, such as reducing pain in arthritic joints, improving gait, and controlling abnormal movements. They can be composed of a variety of materials, such as plastics, metal, and leather. They may or may not have joints to surround and support anatomic joints, such as ankles and knees. Orthoses come in prefabricated varieties but also can be created in a custom fashion to fit the unique needs of any given patient. Some general principles apply to their application with patients. Prescription should be based on careful assessment of the person's particular disorder with a specific goal in mind. The fit should be comfortable with the least amount of restriction to accomplish the purpose(s) of the orthosis. When determining whether an off-the-shelf orthosis vs. custom orthosis is appropriate, the biggest factor is not cost. The best determinants are functional outcome and length of need. If a patient needs the orthosis for only a short period (≤ 6 months), then a noncustom orthosis may be appropriate. However, if the patient has a pathology that is going to require an orthosis for more than 6 months, a custom orthosis is likely the better solution for both fit and function. By placing the needs and functional goals of the patient as a priority and using the clinical expertise of the team members (physician, therapist, and orthotist), a great outcome is possible.⁴¹

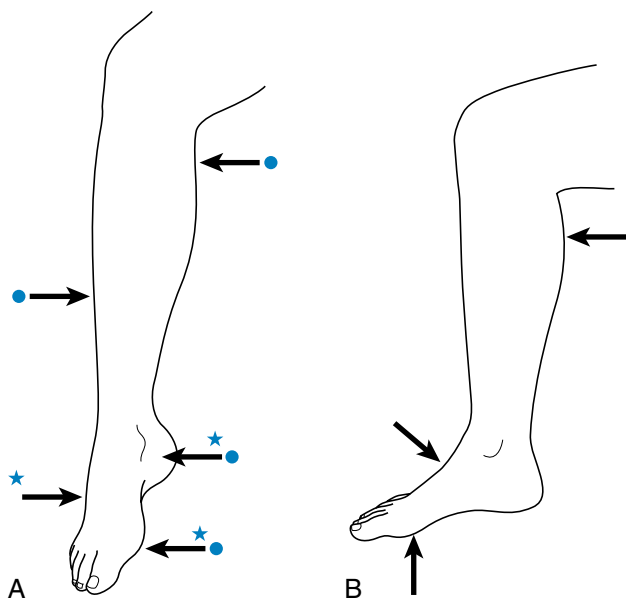
The three-point system of pressure applies to joints such as knees where two points of pressure are applied posteriorly and one anteriorly (Fig. 13.1). An orthotic knee joint should have an axis coincident with the medial femoral condyle, and for the ankle it should be over the tip of the medial malleolus. Hip joints should allow the patient to sit at 90 degrees. Lower limb orthoses should permit good function through all phases of gait.³⁵

Some definitions may be helpful for the subsequent sections. Coxa vara refers to a hip deformity, whereby the angle between the femoral head-neck span and the shaft is reduced to less than 120 degrees. Coxa valga refers to an increase in this angle. Genu varum is commonly referred to as bow-leggedness. With respect to the thigh, the leg angulates medially. Genu valgum is commonly referred to as knock-knee, where the leg angulates laterally. Pes valgus occurs when the medial longitudinal arch collapses, and there is outward turning of the heel. Pes varus is associated with inward turning of the heel. Orthoses are named for the joint they cross over. An ankle-foot orthosis (AFO) crosses the ankle and foot. A knee-ankle-foot orthosis (KAFO) crosses the knee, ankle, and foot. A hip-knee-ankle-foot orthosis (HKAFO) crosses the hip, knee, ankle, and foot.

Shoes

Shoes have been worn for thousands of years to protect the feet from injury and environmental hazards. In the absence of abnormal anatomy or pathology, a shoe is not needed to provide support to the normal foot. A shoe needs to be of the proper fit to prevent skin problems from arising. A callus is a thickening of the skin and/or soft tissue as a result of friction, which can result from shoes that are too loose. A corn is a type of callus that develops as a result of friction over bony prominences, such as the dorsal aspect of the toes, and is seen when a shoe is too tight.

The material used to make a shoe is also important. Leather has traditionally been a good material for most activities because it is pliable enough to conform to the feet as the shoes are worn over time while remaining durable and ventilating well. A pliable sole helps avoid interference with the natural biomechanics of



• **Fig. 13.1** Three-point system control of equinovarus deformity. (A) Control of varus rotational component at the foot (stars) and subtalar joint (dots). (B) Control of equinus deformity. (Modified from Marx HW. Lower limb orthotic designs for the spastic hemiplegic patient. *Orthot Prosthet.* 1974;28:14.)

foot motion. A wide variety of materials are used in the manufacture of modern footwear.

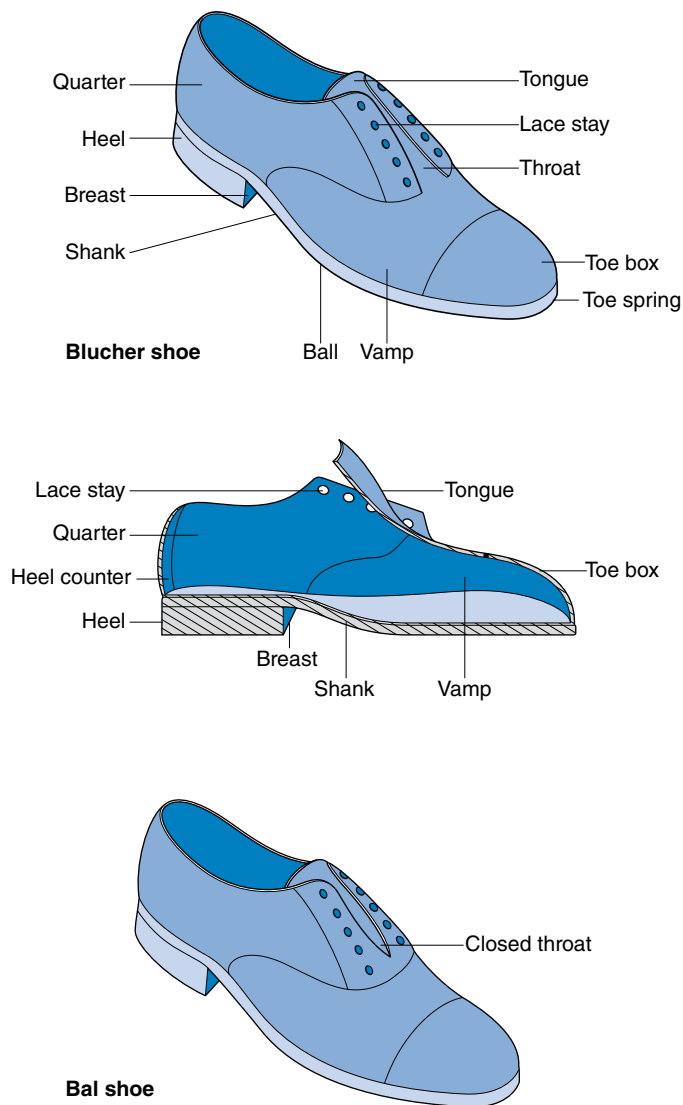
Shoe Parts

Although there are a multitude of shoes on the market, two types of dress shoes, the Blucher and the Bal, are commonly worn (Fig. 13.2). The main distinguishing factor between these two shoe styles is whether the throat is open or not. In the Bal shoe, the quarters (the upper portion of the shoe that wraps around the proximal portion of the foot and allows entry into the shoe) meet at the throat and are stitched to the tongue. In the Blucher shoe, the quarters do not meet, leaving the throat open for more movement of the tongue of the shoe. This allows the shoe to open more and better accommodate the foot in combination with an orthosis.

The toe box is the most anterior portion of the shoe and is followed by the vamp, which holds the forefoot, then followed by the quarters posterior to that. The heel counter is the most posterior portion of the quarters and helps support the back of the shoe, which serves to help control the entirety of the foot (see Fig. 13.2).

Foot Orthoses

There are a wide range of different foot orthoses on the market. They may be bought over the counter or custom manufactured by an orthotist, but whether it is effective in treating its target ailment depends on correctly identifying the underlying problem. Using the most appropriate material and correctly molding the material to the patient's foot are equally important. Over-the-counter orthoses are often sufficient for minor conditions. Custom-molded orthoses may be more effective for certain foot conditions³⁴ and are typically used for more severe or chronic cases. Over-the-counter orthoses are commonly made of soft materials, whereas those custom-molded by an orthotist are more likely to be



• **Fig. 13.2** Shoe types and components. The open throat of the Blucher shoe accommodates an orthosis better than the Bal shoe.

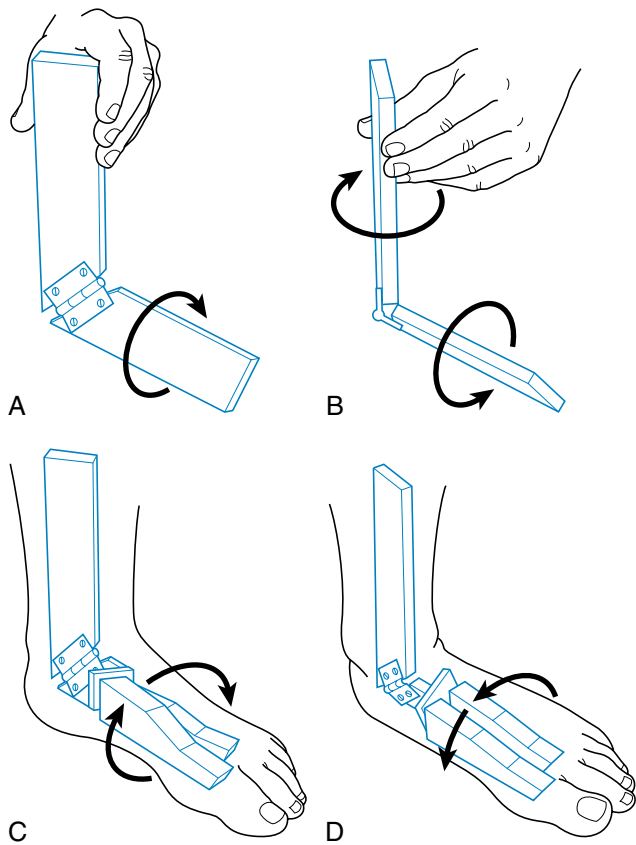
semirigid to provide shock absorption and additional support to the foot. Rigid orthoses are used only for deformities that require aggressive bracing to attain sufficient control.

Foot orthoses work in part by altering the ground reactive forces acting on the joints in the lower limb and by affecting rotational components of gait (Fig. 13.3). Before casting a custom foot orthosis, the subtalar joint should be in a neutral position. Subtalar neutral has traditionally been considered the ideal foot alignment⁷⁶ because it minimizes foot and ankle positional abnormalities, such as hyperpronation. This position helps treat such conditions as pes planus, patellofemoral pain syndrome, and painful rheumatoid arthritis of the first metatarsophalangeal joint.^{33,83}

Common Foot Conditions

Pes Planus

Pes planus, commonly known as flat feet, is exceedingly common, with one study estimating a prevalence at 26.6% in adults.⁷⁵



• **Fig. 13.3** Analogy of subtalar axes to an oblique hinge. (A–C) Outward rotation of the upper stick (tibia) results in inward rotation of the lower stick (calcaneus). This results in elevation of the medial border of the foot and depression of the lateral border. (B, D) Inward rotation of the upper stick (tibia) results in outward rotation of the lower stick (calcaneus). This results in depression of the medial side of the foot with elevation of the lateral side. (Modified from Mann RA. Biomechanics of the foot. In: American Academy of Orthopaedic Surgeons, ed. *Atlas of Orthotics*. Mosby-Yearbook; 1985:118.)

Pes planus is seen with excessive pronation, or in-rolling, of the foot, which manifests as lowering of the medial aspect of the foot and elevating the lateral aspect. Foot pronation is a component of foot eversion. In addition to pronation at the subtalar joint, the motion of eversion also includes dorsiflexion of the ankle joint and forefoot abduction at the tarsometatarsal junction. Pes planus may be caused by increased internal rotation of the tibia (contributing to foot pronation) or abnormal alignment of the calcaneus. The interaction between the tibia and the subtalar joint is what allows nonintrinsic foot abnormalities to cause excessive pronation of the foot.

Excessive pronation of the foot leads to altered distribution of forces and potentially to pain. Pain relief may be achieved by holding the subtalar joint and calcaneus in proper alignment, thereby reducing the degree of pronation. The subtalar joint should be held in a neutral position for the custom-molding process. This prevents rotational deformities present in hyperpronation or supination (see Fig. 13.3C and D).

With the orthosis elevating the anteromedial calcaneus, an upward force is transmitted to the sustentaculum tali, thereby preventing in-rolling.²⁰ By extending the orthosis beyond the metatarsal heads, one obtains increased leverage to control the

foot deformity. The orthosis should also extend proximally to cup the calcaneus and help to control the subtalar joint. A custom-molded foot orthosis of this type, designed to control excessive pronation, is deemed a UCBL, or UCB, orthosis, named after the University of California Biomechanics Laboratory, where this design was researched during the 1940s.

In cases where ligamentous laxity of the foot is causing the pes planus, a medial longitudinal arch support is useful. If the initial arch support is too high, pain may increase, so a stepwise increase in arch height may improve tolerance of the orthosis and patient adherence. Extending the heel medially (known as a Thomas heel extension) may offer additional medial support, especially in patients with a high body mass index.

Useful advice for runners with pes planus or excessive pronation of the foot is to use running shoes with a firm medial heel counter and a wide last at the shank (see Fig. 13.2), both of which serve to reduce pronation at the subtalar joint.

Pes Cavus

In contrast to pes planus, pes cavus is an abnormally high-arched foot, which can cause pain as a result of excess pressure placed on the metatarsal heads. To prevent this, weight bearing should be distributed equally across the metatarsal heads. Additionally, increasing the height of the longitudinal support of the shoe between the shank of the shoe and the arch of the foot serves to better distribute weight to help prevent excess pressure over the heads of the metatarsals. A pes cavus foot tends not to pronate like a pes planus foot, so the arch of the foot is located at the talonavicular joint. A foot may appear high arched if the tibia is externally rotated and the foot is supinated. A custom foot orthosis that holds the subtalar joint in neutral position helps prevent inordinate supination and prevents increased pressure being exerted on the lateral aspect of the foot.

Forefoot Pain (Metatarsalgia)

As mentioned, to help relieve pain on the metatarsal heads, modifications must be made to distribute pressure to other areas. Internal or external modifications can be used to redistribute forces to other areas, typically proximal to the heads of the metatarsal bones. A metatarsal pad is placed inside the shoe just proximal to the heads of the second through fourth metatarsals. A metatarsal bar can also be used when the foot is unable to tolerate a pad within the shoe. This is a wide band of material applied externally to the sole of the shoe just proximal to the metatarsal heads. A rocker bottom shoe helps decrease force on the metatarsals during push-off during the gait cycle by allowing the body's center of mass to passively roll over the base of support rather than requiring active ankle plantarflexion.⁵⁷ Shoes with high heels and/or pointed toes place excessive stress on the metatarsal heads. It is important to educate patients about this fact because selection of appropriate footwear can help to prevent the development of pain.

Heel Pain

Orthoses can help to distribute weight to avoid painful areas of the heel. In relatively minor cases, a heel pad placed inside the shoe can be sufficient. In chronic cases where the foot is unable to tolerate a pad inside the shoe, a calcaneal bar is placed distal to the calcaneus to prevent it from assuming full weight during stance

phase. Alternatively, this can be accomplished using a shoe with a spring for a heel, also set anterior on the calcaneus. Rocker bottom shoes cause heel strike and the subsequent ground reaction force (GRF) to be directed anterior to the calcaneus.

Plantar fasciitis is a disorder of the fibrous tissue along the medial aspect of the sole of the foot that commonly causes medial heel pain where the fascia attaches. Hyperpronation of the foot places increased stress on the medial longitudinal arch. Similar to cases of pes planus, a custom orthosis that holds the subtalar joint in a neutral position serves to reduce pronation and thus stress on the medial arch. The shoe should use a firm medial heel counter with a wide shank.

High arches also predispose to development of plantar fasciitis caused by the increased stress on the medial arch during stance phase. Adding arch support and redistributing pressure along the length of the arch serves to reduce heel pain.

Heel spurs in plantar fasciitis arise from the increased mechanical stress of the plantar fascia on its attachment on the calcaneus. However, the spurs themselves are not necessarily the source of the pain because they are frequently seen in asymptomatic individuals.⁶

Plantar fascia night splints consist of an AFO placed in a few degrees of dorsiflexion.⁸⁰ While the patient is sleeping, the AFO passively stretches the plantar fascia. This can be used for as long as the patient is symptomatic because these splints carry very minimal if any risk, although relief may be seen as early as 1 to 3 months in nonrefractory cases.⁴⁸

Heel lifts serve to decrease tension on the Achilles tendon by keeping the ankle slightly plantar flexed. Achilles enthesitis can be treated with a heel lift used over a period of weeks but should not be used for months because of the risk of developing a plantarflexion contracture. Paradoxically, when plantar flexor spasticity or contracture is already present, a heel lift helps increase the heel height and promote heel strike before toe touch when entering the stance phase during the gait cycle.

Toe Pain

Common causes of toe pain include gout, arthritis, and hallux rigidus (loss of normal mobility of the toe joint caused by arthritis). Orthoses serve to reduce pain from these issues by immobilizing the toes. Reducing mobility of the toe joints may be accomplished through use of a carbon insert extending along the full length of the shoe. If already using a metal AFO, the steel shank may be extended forward to accomplish the same effect.

Leg Length Discrepancy

There are two types of leg length discrepancies: true leg length discrepancy and apparent leg length discrepancy. A true leg length discrepancy may result from an acute trauma, such as a femur fracture, or after a hip or knee replacement surgery. To obtain this measurement, with the patient supine on the exam table, measure from the distal tip of the anterior superior iliac spine to the distal tip of the medial malleolus. An apparent leg length discrepancy may be seen when the true leg length is equal, but anatomic factors such as muscle imbalance, scoliosis, or pelvic fracture create a pelvic obliquity. This measurement is obtained by measuring from a midline landmark, generally either the umbilicus or pubic symphysis, to the distal tip of the medial malleolus. Leg length can also be assessed radiographically.

Ninety percent of the population has some degree of leg length discrepancy, with an average discrepancy of 5.2 mm.⁴⁴ This is often asymptomatic with smaller differences, but a difference in leg length of greater than half an inch is generally considered clinically significant. There is a higher incidence of low back pain when the discrepancy is greater than 12 mm (or approximately half an inch).⁸² If pain improves after a trial of shoe modifications, then the leg length discrepancy can be reasonably assumed to have been a contributing factor.

If the leg length discrepancy is greater than half an inch, the first half inch may be corrected with a heel pad; anything more than this requires external buildup of the heel. Even then, only 75% of the discrepancy should be corrected. The sole of the rest of the shoe must be built up in proportion to the heel.

The Diabetic Foot

The key to treating the diabetic patient who has neuropathy is prevention of any type of foot problems. The biggest cause of foot problems is ill-fitting shoes causing breakdown to the plantar surface of the foot and distal portion of the toes. Impaired sensation and the patient's lack of follow-up and doing frequent skin-foot checks is a recipe for foot ulcer(s) and subsequent amputations.⁹

Diabetic shoes have a higher and wider toe box and an interior surface free of any seams or irregularities. They also have insoles that consist of multilayer, multidense materials that reduce shear forces and pointed pressure areas. For more complex diabetic foot problems, custom shoes and/or custom foot orthoses can be made.

Osteoarthritis of the Knee

Although osteoarthritis of the knee is not a primary foot condition, foot orthoses have been used to alter the GRFs affecting the more proximal joints of the lower limb. For instance, if medial compartment joint space narrowing is present, a lateral heel wedge changes the GRF and attempts to offload the medial compartment of the knee. However, current meta-analyses suggest that this intervention is not effective at reducing medial knee pain caused by osteoarthritis.^{72,94}

Pediatric Shoes

A simple design is best for pediatric footwear. Soft soles allow the feet to develop naturally, and an elevated heel should not be present, so as to facilitate a normal gait. For most children, a tennis shoe is sufficient. A high-cut shoe will stay in place better than a low-cut shoe, especially for very young children. Shoes will need to be changed frequently in the first few years of a child's life because of rapid growth.

Flat feet are commonly seen in infants and children.⁸⁵ They may also occur in asymptomatic adults without the presence of pathology. As subcutaneous fat is lost and joint laxity decreases as the child matures, flat feet tend to improve over time. Overall, current evidence indicates that foot orthoses are beneficial in the treatment of symptomatic pediatric flexible pes planus, but caution should still be exercised because of a lack of standardized parameters and methodological concerns in some studies.¹⁸

Ankle-Foot Orthoses

AFOs are used for problems dealing with foot and ankle issues that are either biomechanical or neurophysiological in nature.

These orthoses include any device that passes across the ankle joint but stays distal to the knee joint. Such orthoses are used for diagnoses that range from ankle sprains to stroke management. This type of brace could potentially be used for any neuromuscular condition requiring support or assistance at the level of the foot and ankle. Of all the various types of lower limb orthoses, AFOs receive the most prescriptions.

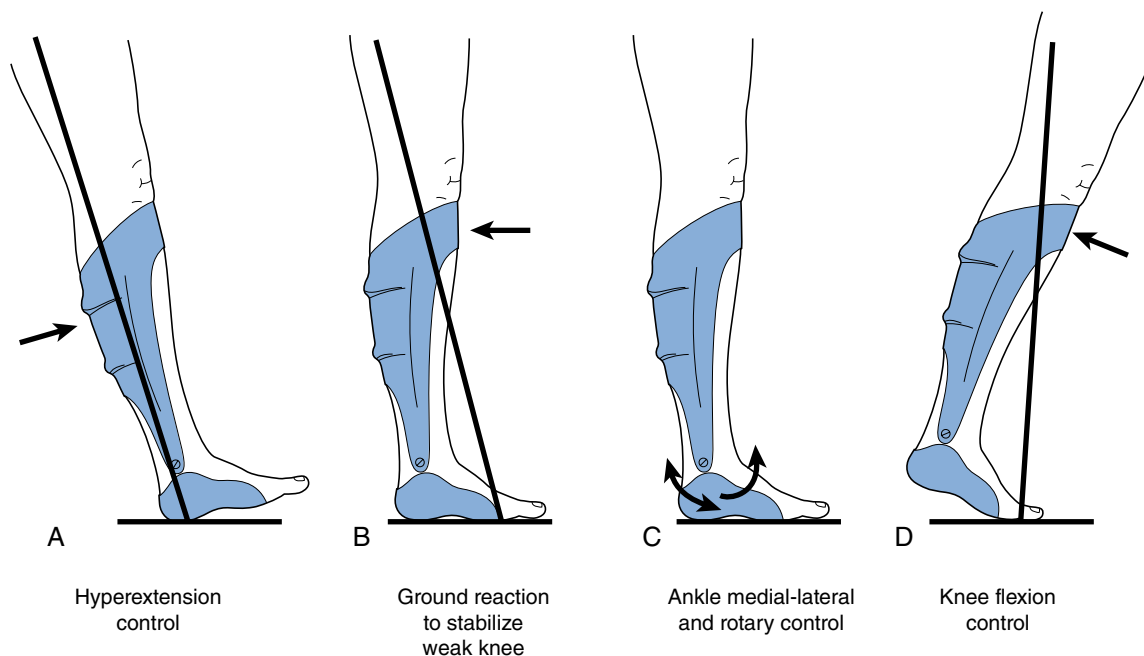
Ankle and foot motions are complex. AFOs provide dorsiflexion and plantarflexion control but also provide mediolateral stability. Motions at the tibiotarsal joint (ankle joint) or subtalar joint and motions of the forefoot at the tarsometatarsal joints combine to provide either eversion or inversion. Eversion consists of ankle dorsiflexion, subtalar pronation, and forefoot abduction. Inversion reverses these motions for ankle plantarflexion, subtalar supination, and forefoot adduction. AFOs provide direct control of ankle motions but also influence control of the knee during gait. Consequently, AFOs can beneficially influence problems with various types of ankle weakness that range from ankle flaccidity to focused areas of weakness, such as with a lack of ankle dorsiflexion that causes foot drag during swing-through in the gait cycle. These orthoses also provide control for spasticity and contractures. Painful ankles, such as from arthritic conditions, can be offloaded to some extent, and wounds such as diabetic plantar wounds can be protected. Examples of conditions that have improved through the incorporation of AFOs include stroke, cerebral palsy, spina bifida/myelodysplasia, mononeuropathies and polyneuropathies, spinal cord injury (SCI) such as in incomplete paraplegia, motor neuron disease in the earlier stages, among others.¹⁴ Gait issues, such as hyperextension of the knee or genu recurvatum, can be beneficially influenced through mechanisms that will subsequently be described.

When the foot strikes the ground, a GRF is exerted against the foot in a vector direction opposite to the striking foot.⁹⁰ The direction of this vector can fall behind the knee, and in this case the GRF will tend to encourage flexion of the knee (Fig. 13.4). If the

GRF falls in front of the knee, then knee extension is assisted. By adjusting certain parameters, an AFO can create either one of these GRFs. Dorsiflexion of the foot tends to create a GRF that is posterior to the knee and enhances knee flexion, while plantarflexion augments knee extension. AFOs with ankle joints can have plantarflexion stops or dorsiflexion stops that encourage the desired action of the GRF on the knee. For example, in someone with weak quadriceps with a tendency to have the knee collapse during gait from weakness, the dorsiflexion stop can be adjusted toward greater plantarflexion during the stance phase. Plantarflexion will move the GRF anterior to the axis of the knee and provide a force to keep the knee more extended. Thus too much flexion of the knee with possible collapse will be discouraged. Conversely, in an individual with a knee that hyperextends during stance, the plantarflexion stop can be adjusted to keep the ankle more dorsiflexed. This adjustment will move the GRF posterior to the knee joint and encourage more flexion and discourage the hyperextension of the knee that was occurring previously. Ideally, these devices would facilitate all aspects of gait; however, in practice, correction of one issue can leave other issues less well attended to. For example, rigid braces designed to protect healing pathology such as fractures or wounds will create more difficulties in walking and require a greater energy expenditure. Another example is the Intrepid Dynamic Exoskeletal Orthosis (Fig. 13.5). This device helps reduce pain and facilitate running for those with limb salvage, but the very features that enhance running make walking more difficult.³⁶

Metal Ankle-Foot Orthosis

Metal AFOs are composed of metal uprights, a foot plate or other interface between the brace and the shoe, and leather straps (Fig. 13.6). The ankle joints of these types of braces have pins or springs for adjustments that are adapted to various pathologies and needs (Fig. 13.7). The metal in the posterior area of the calf band



• **Fig. 13.4** (A–D) Ground reaction ankle-foot orthosis (AFO) dynamic illustration. Note the effect of the proximal portion of the AFO on the knee throughout gait. (Courtesy Oregon Orthotic System, Albany, OR.)



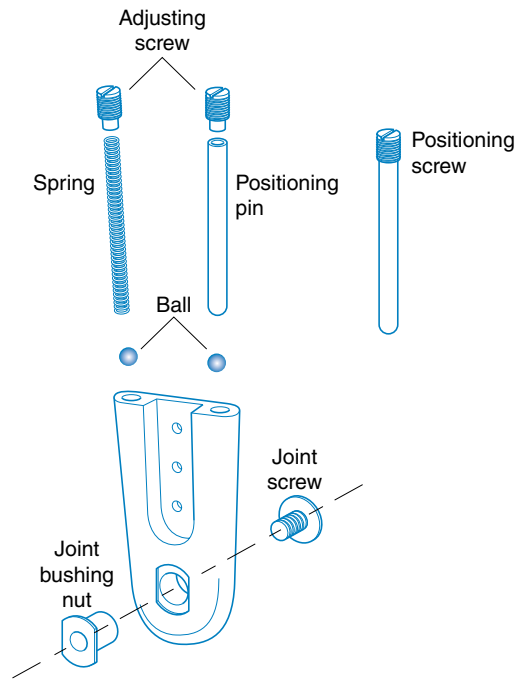
• **Fig. 13.5** Intrepid Dynamic Exoskeletal Orthosis (IDEO) custom ankle-foot orthosis. (From Fox JR, Lovegreen W. Lower limb orthoses. In: Webster JB, Murphy DP, eds. *Atlas of Orthoses and Assistive Devices*. 5th ed. Elsevier; 2019:239-246.)



• **Fig. 13.6** Double-action metal ankle joint with solid stirrup.

should be between 1.5 and 3 inches for enough weight distribution to allow for comfort and prevent skin irritation or breakdown. The top of the calf band should rest at least 1 inch below the fibular neck to avoid compression on the peroneal nerve.⁶⁴

The interface with the shoe consists of either a plate (see Fig. 13.6) with medial and lateral uprights for the ankle joints or a split stirrup that fits into the side of the shoe. The single plate is fixed firmly to the shoe and can extend to the metatarsal heads



• **Fig. 13.7** Schematic drawing of metal ankle joint components. This type of ankle joint is most commonly referred to as a double-action ankle joint. It also has been referred to in the distant past as a double Klenzak ankle joint or a bichannel adjustable ankle locking (BICAAL) joint. (Courtesy USMC, Pasadena, CA.)



• **Fig. 13.8** Split stirrup. The stirrup extends anteriorly to attach at the shank area of a shoe for stability. This stirrup is for dorsiflexion assist.

or further. The split stirrup (Fig. 13.8) can be adapted to fit into different shoes but lacks the support provided by the single plate.

The pins on the joints can be tightened or loosened to provide restriction to dorsiflexion or plantarflexion. There are anterior and posterior stops on the medial and lateral joints. The anterior stop limits dorsiflexion, while the posterior one limits plantarflexion. Straps can be attached near the joints to limit either a tendency toward varus or valgus. A medial T strap will apply pressure over the medial malleolus and buckle around the lateral upright. A lateral T strap will apply pressure over the lateral malleolus and buckle around the medial upright.

In general, metal AFOs have been replaced by plastic and other materials, but metal AFOs still have utility and are prescribed in certain circumstances. For example, an individual may have

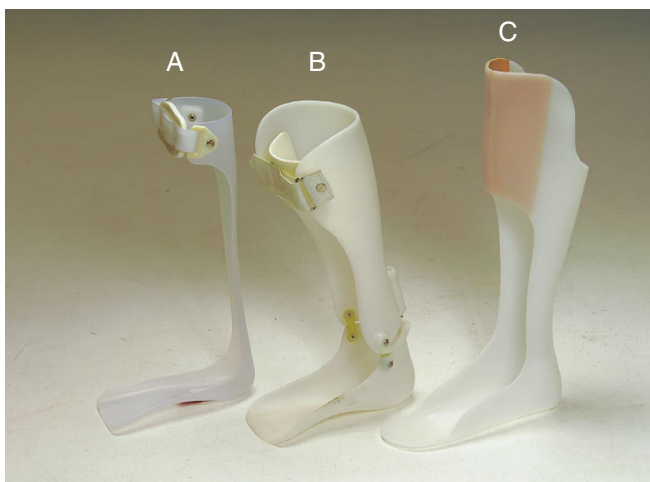
grown accustomed to a metal brace and thus despite the advantages of newer types may not want to change. In other instances, the weight of an individual may require the sturdiness that a metal AFO can provide. Skin conditions such as sores, weeping, or exudative lesions can necessitate the use of metal AFOs as well.

Plastic Ankle-Foot Orthosis

Plastic AFOs (Fig. 13.9) come in a wide variety of styles and are composed of different types of materials. Examples of plastic AFOs are the solid ankle, posterior leaf spring, hinged, spiral, hemispiral, Swedish, energy return, and short-leg fixed AFOs. Most plastic AFOs are made of polypropylene. Some are hybrids of this material and a canvas fabric. A lesser number of orthoses are made with copolymer. Soft foam or plastic can be combined with a harder, more durable shell. These types of devices may or may not include medial and lateral joints with anterior and posterior stops similar in function to those mentioned for metal AFOs. These joints can be single or double action and may have spring-assisted dorsiflexion to assist with toe clearance during swing phase. The position of trimlines either around the ankle or the foot can be varied as necessary so that if more support is necessary the trimline can be extended. The length of the foot plate can vary from proximal to distal to the metatarsal heads.

Other Types of Ankle-Foot Orthoses

Although plastic AFOs are the most common, other variations have been tried and are in use. Examples include carbon fiber and laminated AFOs. The laminated AFOs are made in a way that is similar to the way laminated sockets are made for prostheses. Hybrids have been created, such as plastic and carbon fiber, plastic and metal, and carbon fiber and metal. One example is the Blue Rocker AFO⁸¹ (Fig. 13.10), which is a carbon composite. This is designed for foot drop with associated flaccidity or varying levels of spasticity. It can be used with partial foot amputations as well. Fior & Gentz makes the Neuro Swing AFO⁶⁴ that allows for adjustment of three parameters in a single ankle joint to facilitate the gait in individuals with neurological disorders. The patellar tendon-bearing orthosis is designed to offload structures such as the ankle joint that are distal to the pressure-bearing areas such as the



• **Fig. 13.9** Common ankle-foot orthosis (AFO) prescriptions. (A) Foot drop AFO. (B) Plantar spasticity AFO. (C) Lumbar spinal cord injury AFO.



• **Fig. 13.10** Blue Rocker noncustom ankle-foot orthosis, Allard Inc. (Courtesy of CAMP Scandinavia AB.)



• **Fig. 13.11** The Charcot restraint orthotic walker boot is a custom-molded bivalve plastic ankle-foot orthosis that is fully padded on the inside with total contact of the foot, ankle, and leg. The purpose of this brace is to offload a plantar ulcer or stabilize the progressive deformity from Charcot joint of the foot and ankle.

patellar tendon, tibial condyles, and soft structures that are bearing weight proximally. Indications for this orthosis include Charcot joint, ankle joint fusions in the postoperative stages, calcaneal fractures, diabetic heel ulcers, and avascular necrosis of the foot or ankle. Other conditions include nonunion and delayed union of fractures, arthritis of the joints in the foot or ankle, and soft tissue issues such as loss of the heel pad. This type of AFO allows the GRFs to at least partially bypass the pathological areas and be transmitted to more tolerant structures proximally.

The Charcot restraint orthotic walker⁴⁵ (Fig. 13.11) provides a firm support from a plastic, custom-molded bivalve orthosis or

clamshell design with a soft interior and a rocker bottom of rubber. Persons with a Charcot joint can wear this to impede progression of their condition to further deformity. The orthosis can also provide protection for a plantar ulcer complicated by diabetes and can assist in the management of insensate feet. The device is not worn with any other footwear.

Patients with limited mobility, such as those with hemiplegia, SCI, and coma, often require an orthosis to prevent permanent contractures from developing in the ankle and to offload the heel and the malleoli. Brand name orthoses that provide this type of protection include the Multi-Podus and the PRAFO, in which the heel is free floating. In addition, to fully prevent contractures, the ankle should receive range of motion (ROM) exercises to complement the action of the orthosis. Controlled ankle movement walker boots permit walking through preventing any motion at the ankle or allowing a very limited amount in individuals with extensive injuries, such as fractures, sprains, and tendon tears.

Electrical Stimulation Devices

Because foot drop can occur with a loss of signal from the peroneal nerve to the muscles that dorsiflex the foot, devices have been created (WalkAide [Fig. 13.12] and L300 from Bioness, Inc.) that

provide an electrical stimulus to nerves and muscles at the appropriate time in the gait cycle to activate dorsiflexors, improving toe clearance.²² A sensor in the shoe can communicate the gait information to the stimulator. The user can adjust the level of stimulation through a handheld device. Stroke, incomplete SCI, traumatic brain injury, and multiple sclerosis represent conditions in which these devices could prove useful.

Case Report

The following case report⁸ illustrates the benefits and effects of an appropriate orthosis for a given condition. A ToeOFF AFO was prescribed for a 42-year-old left-handed male with right hemiplegia caused by a left middle cerebral artery stroke. Cadence, stride length, and step length all increased for a net increase in walking speed. At the same time, the energy cost of walking decreased. The hip kinematics showed significant changes with increased hip flexion bilaterally at foot strike, and the angular velocity of the hip on the right increased during swing phase. Hip circumduction decreased, and the biomechanical position of the leg improved. As this case illustrates, provision of the appropriate orthosis can provide benefits in multiple ways through appropriate adjustments in the biomechanics of gait.⁶⁹



• **Fig. 13.12 (A–D)** WalkAide functional electrical stimulation system. (Courtesy Innovative Neurotronics, Austin, TX.)

Knee Orthoses

General Considerations

Knee orthoses (KOs) are a very commonly prescribed class of brace both because of the high prevalence of knee osteoarthritis and the high rate of knee injuries in recreational and professional athletes.^{1,17,32,50,61} KOs are designed to provide stability, prevent instability, limit motion, and/or provide offloading of the knee joint. Most include both a medial and lateral upright with either a free motion or adjustable knee joint. A KO is primarily indicated when the ankle and foot are fully functional, such that a KAFO is not required. The indications for provision of a KO are further delineated in [Box 13.1](#).

Brace Design Features

Although many of the KOs have similar features, subtle differences in the materials used and design features have the potential to impact the fit, comfort, and function of the brace.^{5,55} [Box 13.2](#) highlights the important KO design features that should be taken into consideration to provide the most appropriate brace. Most KOs being prescribed are prefabricated-type braces. There are also a host of prefabricated KOs available commercially. Many of these off-the-shelf devices have some degree of adjustability for customization for the individual. Fully custom KOs are less frequently prescribed and used.

KOs have traditionally been fabricated with metal, plastic, and fabric materials. The materials used in sleeve-type KOs have evolved to provide enhanced compression, comfort, and support; improve donning; and reduce skin complications ([Fig. 13.13](#)).^{29,55,62,84} Braces have been increasingly manufactured with lightweight composite materials, such as carbon fiber and aluminum alloy.^{5,29,62,84}

• BOX 13.1 Clinical Indications for Knee Orthoses

Medial/lateral knee instability (medial/lateral collateral ligament strain or tear)
Anterior/posterior knee instability (anterior/posterior cruciate ligament strain or tear)
Postoperative stabilization or support
Prophylactic bracing to reduce the risk of ligamentous injury
Motion restriction (flexion and/or extension/hyperextension, rotational control)
Offloading (medial or lateral compartment osteoarthritis)
Patellar support and/or control of patellofemoral tracking

• BOX 13.2 Knee Orthosis Design Feature Considerations

Prefabricated vs. custom-fit vs. custom
Uprights (none, unilateral, bilateral)
Straps (width, location, padding, materials)
Joint (single axis or polycentric, locking or free motion)
Range-of-motion restriction (no restriction, set restriction, adjustable restriction)
Donning method (pull-on, wrap around, step-through)
Suspension mechanism
Materials (metal, plastic, carbon-composite)



• **Fig. 13.13** Standard knee sleeve with patellar cut-out. (Courtesy Medline Industries, Inc.)

This has decreased the weight of the brace while maintaining durability and strength. Some brace fabrication concepts also take into consideration the importance of the skin interface with padding or foam inserts. The use of undersleeves or silicone liners can improve comfort and reduce migration, which helps with compliance.

Important considerations are the mechanism and materials used for suspension of the orthosis. Suspension of a KO can be challenging, especially for individuals with conical-shaped thighs or excessive soft tissues. Suspension relies upon the use of contoured uprights in conjunction with correctly placed strapping. Adequate sizing and appropriate tightening of straps are also important to decrease the potential for slippage. If needed in extreme conditions, an AFO section can be attached to the KO for enhanced suspension and alignment.

Prescription Considerations

Prescription is complicated by the wide variety of KOs that are commercially available from a large number of different manufacturers. KOs range from simple pull-on knee sleeves to sophisticated braces with adjustable ROM limitations and offloading capabilities.^{5,29} Different classification schemes have been used to classify KOs. These classification schemes may take into consideration whether the orthosis is prefabricated, custom, or custom fit. Other classification schemes are based on whether the brace is designed to be supportive, corrective, or functional. Although some braces are designed to primarily serve one purpose, others have features that address several different functions. [Box 13.3](#) lists some of the most commonly prescribed types of KOs.

Simple knee immobilizers have no joint and are designed to keep the knee in full extension after knee injury or surgical intervention. Sleeve-type KOs can be simple in design and provide mild compressive support to the knee, whereas others can incorporate articulated hinges for greater medial-lateral support.⁶² Other sleeves are designed to specifically provide

• BOX 13.3 Commonly Prescribed Types of Knee Orthoses

Knee immobilizer
Knee sleeve
Swedish knee cage
Prophylactic sports brace
Patellar support brace
Supportive bracing for ligamentous instability
Unloader braces for medial/lateral compartment unloading



• **Fig. 13.14** T Scope Premier double upright knee orthosis with range-of-motion restriction feature. (Courtesy Breg, Carlsbad, CA.)

patellofemoral joint support and corrective forces for patellofemoral tracking conditions.^{15,25,49,84} Prophylactic braces can be designed with unilateral, single-hinge systems or with bilateral uprights. These braces typically allow for free knee flexion and extension, but other models are available that restrict motion.^{31,63,65} This feature can also be used as an individual is gradually advancing through a rehabilitation program after an injury or surgery (Fig. 13.14).

Functional knee braces typically are designed with bilateral uprights with either single axis or polycentric joints (Fig. 13.15).^{46,47,56,78} The KO known as a Swedish knee cage is used to control minor to moderate genu recurvatum caused by ligamentous or capsular laxity. Offloading KOs are most commonly prescribed for individuals with osteoarthritis where either the medial or lateral knee compartments are more severely affected.^{5,11,17,29} These designs use a traditional three-point system to create either a varus or valgus stress on the knee joint to preferentially offload either the medial or lateral compartment (Fig. 13.16).

Despite being very commonly prescribed and widely available commercially, the effectiveness of these KOs remains debatable, and accurate, unbiased information regarding the proper use and outcomes associated with these devices is limited.^{5,29,55,62,77,79,84}



• **Fig. 13.15** Townsend Premier double-hinged functional knee brace for postinjury or postsurgery stabilization of ligament instabilities. (Courtesy Townsend Design. http://www.townsenddesign.com/Townsend_Design/Premier.html.)



• **Fig. 13.16** Knee orthosis with three-point distribution (arrows) for unloading of the medial or lateral knee compartments.

There is a great deal of controversy particularly surrounding the use of prophylactic knee braces in sport.^{31,63,65} The lack of studies demonstrating comparative efficacy between different designs and unique features can leave players, coaches, therapists, and other health care professionals with little evidence to support clinical decision making.

Knee-Ankle-Foot Orthoses

General Considerations

KAFOs are designed to provide stability to the knee and ankle complex during ambulation. The KAFO may be required to provide stability during the loading response of the gait cycle by preventing excessive knee flexion or during midstance to terminal stance by preventing knee hyperextension. Ideally, the KAFO provides the required stability while minimally interfering with normal knee and ankle ROM during the gait cycle. KAFOs can be prescribed and used for functional ambulation, exercise, or both.^{58,59} For some individuals, the use of KAFOs is complemented by the use of a wheelchair for longer distance mobility. Good trunk control and upper body strength are typically required to ambulate with KAFOs because these devices are used in combination with such ambulation aids as walkers or forearm crutches.

Design Feature Considerations

Ankle Joints

Any style of ankle joint or ankle joint system available for use in AFOs (as previously described) can be incorporated into KAFOs.^{27,58,59} KAFOs most commonly use a double upright design at the ankle, but a unilateral upright or a posterior strut without an ankle articulation can also be used. Double uprights attached with stirrups to the outside of the shoe accommodate edema and can be used with sensory impairment, but these styles can be heavy and make switching shoes difficult. Molded plastic ankle joints with or without an articulation allow the patient to switch shoes more easily and potentially provide more detailed control of the ankle and foot complex (Fig. 13.17).



• **Fig. 13.17** Knee-ankle-foot orthosis with articulated thermoplastic ankle-foot orthosis component and posterior offset knee joint. (From Lovegreen W, Kwasniewski M, Panchang P, et al. Orthotic management of polio and postpolio syndrome. In: Webster JB, Murphy DP, eds. *Atlas of Orthoses and Assistive Devices*. 5th ed. Elsevier; 2019:277-288.)

Knee Joints

There are three traditional types of knee joints.^{56,58,89} The straight, single-axis knee joint provides rotation about a single axis. It allows free flexion but prevents knee hyperextension. Free-swinging knee joints are indicated when medial-lateral support is required, but adequate knee extensor strength is present. This type of knee joint can be used in combination with a drop lock if needed to keep the knee locked in extension during all phases of gait. The polycentric knee joint uses a double-axis system to simulate the flexion-extension and gliding motion of the femur across the tibia. Although this type of joint provides more physiological motion, it is not used as commonly as the other types of joints. The third type of knee joint is the posterior offset knee joint (Fig. 13.18). This joint allows free flexion and extension of the knee during the swing phase of gait. During stance phase, the posterior offset of the joint helps keep the orthotic GRF in front of the knee axis for stability. The posterior offset knee joint should have a hyperextension stop to help prevent genu recurvatum.

Knee Locks

Although the knee joint on a KAFO can be designed to be free swinging, knee locks are used to provide enhanced stability as indicated for the individual patient. The four traditional types of knee locks are the drop lock, ratchet lock, dial lock, and bail lock. More recently, various types of stance control and dynamic control locking mechanisms have been introduced. Stance control lock mechanisms are activated to lock during stance phase but allow knee motion (flexion and extension) to occur during swing phase.^{27,58,89} Dynamic systems provide some level of knee motion control during both the stance and swing phases of gait through mechanisms such as pneumatic or hydraulic systems.⁸⁹

The drop lock provides a simple and lightweight design; however, fine motor coordination in the hands is required to lock and unlock the knee. Another disadvantage of the drop lock is that there is no locking mechanism until full knee extension is



• **Fig. 13.18** Fillauer stance control knee-ankle-foot orthosis with mechanical knee control mechanism. (Courtesy Fillauer, LLC, Chattanooga, TN.)

obtained. A drop lock can be used with a single upright if the patient is relatively lightweight and has a low activity level.

The ratchet lock has a catching mechanism that operates in 12-degree increments. As the user rises from a seated position, the ratchet lock prevents instability by blocking flexion and keeping the gains made toward extension. With the knee extended, knee flexion is achieved either by pressing down on a release lever or by sliding the locking mechanism open.

The dial lock is also used to stabilize the knee in varying amounts of flexion. It can be adjusted in 6-degree increments and is more precise than the ratchet lock for the management of a knee flexion contracture. Both the ratchet and the dial lock systems can be used to prevent progression of a flexion contracture or assist with the gradual reduction of a flexion contracture.

The bail lock provides an easier method of simultaneously unlocking the medial and lateral knee joints of a KAFO. Lifting the posteriorly located bail mechanism releases the knee joint to permit flexion, allowing the patient to sit down. The patient can also catch the bail on the edge of a chair to release the lock mechanism to permit sitting. The locking mechanism is spring loaded to assist locking the knee into full extension.

Stance control KAFOs use different mechanisms to lock and unlock the knee joint during the gait cycle. Some of these mechanisms are weight activated, whereas others use different gait parameters, such as hip extension or ankle dorsiflexion, to release the lock mechanism.^{71,89,92} These systems most commonly use full extension of the knee joint to initiate locking of the joint. The advantage of weight-activated systems is that some will lock in place to prevent knee flexion during weight bearing without full knee extension being achieved. Some of these stance control locking mechanisms are mechanical in nature, whereas others use magnetic or electronic switches (Fig. 13.19).⁶⁹ More advanced technologies use dynamic control of knee resistance throughout the gait cycle. The C-brace (Otto Bock) incorporates a microprocessor control system with a hydraulic joint. The hydraulics and microprocessor control allow flexion and extension dampening in both swing and

stance phases. The microprocessor control adjusts quickly to varying gait situations and sensor feedback, allowing the individual to more easily traverse inclines and stairs.⁸⁹ The C-brace also has a standing feature, which keeps the knee locked for prolonged standing. The benefits of stance control and dynamic knee control units include greater stability on uneven terrain and reduced metabolic costs of ambulation. Reducing gait deviations, such as hip hiking and circumduction, may also lead to a decrease in longer term complications in the back and contralateral limb.

Additional Component and Material Considerations

The materials selected for use in the KAFO will influence the strength, durability, flexibility, and weight of the brace. Traditionally, the materials used in KAFOs were metal (primarily steel) and leather. Although strong and durable, braces using these materials are heavy.^{27,58,59,89} Currently, most KAFOs are made of more lightweight components consisting of aluminum, thermoplastics, and carbon fiber composite materials in combination with synthetic fibers and Velcro straps.^{71,89,92} Bands and/or shell structures are used at the thigh and the calf to secure the brace to the person and provide the control points for knee stability. The thigh band needs to be wide enough to adequately distribute the pressure of the forces transmitted through the knee axis. A partial thigh shell can provide a larger contact area and reduce the risk of high-pressure areas on the skin. An additional thigh band located low on the thigh may be indicated to control genu recurvatum. In a similar manner, the calf component of the KAFO can consist of a padded band or a molded calf shell. Foam liners can be used to increase comfort with shell style braces.

Clinical Indications

As described, the purpose of a KAFO is to provide stability at the knee, ankle, and subtalar joints during ambulation. Clinical indications for a KAFO include lower extremity weakness involving the knee extensors and flexors, structural knee instability, varus or valgus deformity at the knee, knee flexion spasticity, or hyperextension of the knee (Box 13.4).^{10,71} One of the biomechanical problems with traditional KAFOs that maintain full knee extension during gait is the effect it has on the normal gait pattern. Providing a person who has a very unstable knee during stance phase with a locked knee allows the person to stand and ambulate safely. However, having a locked knee during the swing phase creates a need for hip hiking, circumduction of the limb, and/or vaulting in on the contralateral side to advance the limb forward. These gait deviations can result in long-term musculoskeletal complications. Advances in stance control and dynamic knee



• **Fig. 13.19** E-Mag stance control knee-ankle-foot orthosis with electromagnetic knee control mechanism. (Courtesy Ottobock, Duderstadt, Germany.)

• BOX 13.4 Clinical Indications for Knee-Ankle-Foot Orthoses

- Upper motor neuron lower extremity weakness (stroke, brain injury, spinal cord injury)
- Lower motor neuron lower extremity weakness (polio, peripheral neuropathy, myopathy)
- Lower extremity weakness and any of these associated conditions:
 - Intrinsic knee joint instability
 - Knee hyperextension
 - Varus or valgus knee deformity
 - Impaired motor control and hypertonicity
 - Knee pain
 - Sensory impairment

control mechanisms, including the use of microprocessor control, have allowed patients to return to more normal gait patterns with reduced metabolic costs.

Hip Orthoses

Hip orthoses are used to provide support for patients with hip disorders or after hip surgeries. They can provide control in various planes of hip motion, including flexion-extension, abduction-adduction, and internal-external rotation. They may be unilateral or bilateral. In cases where prefabricated orthoses are not sufficient, they may be custom molded to fit the patient. Common uses for hip orthoses include treating hip dislocation, hip dysplasia, or after hip replacement surgery (hip arthroplasty). By helping to position the femoral head in the acetabulum, they allow for healing or prevention of further damage to the joint. The hip orthoses currently in use have hip joints that use an adjustable multipositional lock for flexion-extension control, abduction-adduction control, or both (Fig. 13.20). If the pelvic and/or thigh section of the orthosis are loose or ill fitting, attempts to control a dislocating hip prosthesis are likely to fail. For this reason, ensuring a proper and complete fit is of utmost importance.

Hip-Knee-Ankle-Foot Orthoses

HKAFOs are normally KAFOs (usually bilateral) that are attached to a hip device (pelvic band, lumbar sacral orthoses [LSOs], or thoracic LSOs [TLSOs]) for medical conditions warranting hip control. Box 13.5 provides a list of possible indications for



• **Fig. 13.20** Newport MC hip orthosis. (Courtesy Orthomerica, Orlando, FL.)

• BOX 13.5 Possible Indications for Hip-Knee-Ankle-Foot Orthoses

Lower thoracic/upper lumbar paraplegia
Spina bifida
Muscular dystrophy
Rotational control

HKAFOs. Paralysis of hip abductor muscles is one of the most common reasons for prescribing HKAFOs. HKAFOs are usually fabricated using mechanical hip joints, most commonly made of metal. They can incorporate flexion-extension and abduction-adduction control and have free or locking joints. There are many types of HKAFOs that range from the conventional style of metal and leather components attached to shoes to the more complicated total contact, molded plastic reciprocating gait orthosis (Figs. 13.21 and 13.22).⁴¹



• **Fig. 13.21** Standard hip-knee-ankle-foot orthosis. (From Fox JR, Lovegreen W. Lower limb orthoses. In: Webster JB, Murphy DP, eds. *Atlas of Orthoses and Assistive Devices*. 5th ed. Elsevier; 2019:239-246.)



• **Fig. 13.22** Reciprocating gait orthosis. (From Fox JR, Lovegreen W. Lower limb orthoses. In: Webster JB, Murphy DP, eds. *Atlas of Orthoses and Assistive Devices*. 5th ed. Elsevier; 2019:239-246.)



• **Fig. 13.23** PRAFO ankle-foot orthosis. (Courtesy Anatomical Concepts, Inc., Poland, OH.)



• **Fig. 13.25** Orlando TPFx. (Courtesy Orthomerica, Orlando, FL.)



• **Fig. 13.24** Shark-O Charcot orthosis. (Courtesy Orthomerica, Orlando, FL.)

Specialty Lower Extremity Orthoses

Additional lower extremity orthoses include the positional orthoses that are used to prevent and decrease contractures (Fig. 13.23), prevent heel ulcers (see Fig. 13.23), provide treatment of a Charcot diabetic foot (Fig. 13.24), assist fracture healing (fracture orthoses) (Fig. 13.25), and enhance postoperative care (temporary postoperative orthoses) (Fig. 13.26).

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Orthotic Checkout

There are multiple considerations when checking out the appropriateness of an orthosis that has been ordered. The overall goals



• **Fig. 13.26** DAFO 9. (Courtesy Orthomerica, Orlando, FL.)

of the checkout are to see if the device is acceptable in terms of fit, function, comfort, and cosmesis. For starters, one should determine if the orthosis matches the original prescription. An examination of the patient will establish if any changes have occurred that would affect the fit or function of the orthosis, such as weight gain or edema of the limb. The orthosis should be examined in both static and dynamic situations. It should fit comfortably. The ankle joint should coincide with the tip of the medial malleolus. The screws should be tight and trim lines smooth. If the orthosis was prescribed to improve a gait dysfunction, gait should be reassessed to ensure the orthosis accomplishes that goal. For example,

if the orthosis was intended to correct a foot drop during the swing phase of gait, there should be no foot drop. The proximal edge of an AFO should be at least 1 inch below the fibular head. The skin should be examined after wear to ensure that no red areas last more than 5 to 10 minutes and that there is no skin breakdown. The patient should find the orthosis acceptable and feel that it is something that they will continue to wear.

Powered Exoskeletons

Powered exoskeletons provide the potential for individuals with various levels of SCI and similar conditions to achieve ambulation in the hospital, home, and community settings (Video 50.1).³⁰ Powered exoskeletons are designated as Class II (intermediate risk) medical devices by the US Food and Drug Administration (FDA). Various brands of powered exoskeletons have become commercially available, and several are FDA approved for use in different environments. To date, three of the four commercially available powered exoskeletons have received FDA marketing clearance for use in persons with SCI (ReWalk in 2014, Ekso and Indego in 2016) (Fig. 13.27). The various brands serve a range of SCI levels. The Ekso GT (Ekso Bionics, California, USA) and the REX (REX Bionics, London, UK) are able to be used in individuals with cervical level injuries. Others, such as the ReWalk (ReWalk Robotics Inc., Yokneam Illit, Israel) and Indego (Parker Hannifin, Ohio, USA; indego.com), may be applicable for those with injuries below the T4 level. Triceps function is needed for crutch stability for all except the REX.³⁰ The ReWalk was the first system to be cleared by the FDA for both personal and rehabilitation use in the United States. Recently, this company has developed a unilateral version that provides support and muscle reeducation and strengthening for post cerebrovascular accident hemiparesis rehabilitation.

A host of benefits have been described with the use of powered exoskeletons in individuals with SCI.^{30,60,68,93} In selected instances, these exoskeletal devices allow previously nonambulatory individuals to ambulate either independently or with supervision.



• **Fig. 13.27** ReWalk powered exoskeleton. (Courtesy ReWalk Robotics, Marlborough, MA.)

Achieving ambulation in individuals with SCI improves their level of physical activity and cardiovascular fitness, and research is ongoing about potential short- and long-term health benefits that powered exoskeleton devices might impart. This technology has also demonstrated an impact on psychological well-being by improving quality of life and decreasing anxiety and depressive symptoms. Maximizing the benefits of these devices is accomplished by combining the device use with other established techniques, including exercise, gait training, and functional electrical stimulation.

Ambulation Aids

A variety of ambulation aids are available to increase the area of support during standing and ambulation based on the individual needs of the patient.^{72,75} The type of aid provided should be based on the severity of the impairment and the extent of difficulty the patient has maintaining a center of gravity safely over the support area. Ambulation aids have the potential to enhance balance, widen the base of support, reduce load on one or both lower limbs, provide sensory feedback cues, and potentially facilitate propulsion.^{3,4,8} These aids act as an extension of the upper limb, and their proper use generally requires adequate upper limb strength and coordination. In addition, ambulation aids may allow the user to experience the physiological benefits of upright posture and walking, maneuver where a wheelchair cannot fit, and alert others to the special needs of the user.^{23,24}

Canes

Indications

Prescription of a cane is indicated for individuals with mild-moderate balance impairment and for those who will benefit from a mild to moderate reduction in weight bearing on an affected limb.^{26,39,40}

Types of Devices

There are several different types of canes (Fig. 13.28). The single-point cane has a single tip that makes contact with the ground.



• **Fig. 13.28** Various ambulation aids. *Left to right*, Cane, forearm crutch, standard walker, axillary crutch, and quad cane.

Several different handle styles are available, including the C-shaped handle and the functional flat grip handle. A quad cane has four points of contact with the ground and provides an increased area of support compared with a single-point cane. Quad canes come in narrow- and wide-based forms for different degrees of support, with the wide-based style being indicated for those with more significant balance impairment or need for a greater amount of weight offloading. A visual impairment cane is a specialized type of cane with a longer shaft and red coloration on the distal shaft that provides sensory feedback to the user through both sound and vibration.^{23,24}

Fitting and Use Considerations

The cane should be measured to the level of the greater trochanter with the patient in an upright position. The elbow should be flexed approximately 20 degrees. A cane is used on the side opposite the lower limb requiring the support or offloading. During ambulation, it is advanced with the opposite lower limb.

Crutches

Indications

The prescription of crutches is generally indicated for individuals with mild to moderate balance impairment and for those who require complete offloading of a limb because of an injury or illness. Various styles of crutches are available, with the most common types mentioned next.

Types of Crutches

- **Axillary:** Crutch type that provides support from the axilla to the floor; however, these crutches are not designed for weight bearing through the axilla. There are several different styles of axillary crutches with varying upper segments to improve weight-bearing distribution and function.
- **Forearm:** Crutch type with a single tip in contact with the floor and a bent proximal shaft with one or two cuffs that encircle and support the forearm. The forearm cuffs should pivot to allow the patient to lean on the crutch for free use of the hands.
- **Triceps:** Crutch type with two posterior bands that help to mimic the action of the triceps muscle and maintain the elbow in an extended position.
- **Platform:** Crutch type that incorporates a platform trough that permits weight bearing through the forearm for those who are unable to tolerate weight transmission through the hand.

Fitting and Use Considerations

Various gait patterns can be used with different crutch types depending on the impairment and the extent of limb offloading that is required.^{28,38} Two- or four-point gait patterns can be used when the balance impairment is mild to moderate and when complete offloading of the limb is not required. A three-point gait pattern can be used to provide greater to full offloading of a single limb. A swing-through gait pattern can be used when both limbs are affected and braced or during complete offloading of one limb. The length of the axillary crutch should be measured as the distance from the anterior axillary fold to a point 6 inches lateral to the fifth toe with the patient standing with the shoulders relaxed. The hand piece should be measured with the patient's elbow flexed 30 degrees, the wrist in maximal extension, and the fingers forming a fist. Patients using axillary crutches should be reminded

to not allow direct pressure or weight bearing through the axilla to avoid compressive radial neuropathies or plexopathies.

Walker

Indications

The prescription of a walker is indicated for individuals with moderate to severe balance impairment and for those who will benefit from a moderate to high level of weight-bearing reduction on an affected limb.^{23,24,42,87}

Types of Walkers

- **Hemiwalker:** This walker type is indicated for those who require the stability of a walker but have unilateral weakness or other upper extremity impairments that prevent bilateral control of the walker.
- **Standard:** This walker type is indicated for those who require the greatest amount of stability and support for ambulation. This walker type requires the user to be able to lift the walker to advance forward. Gait is slowed with the use of a step-to gait pattern.
- **Front wheeled:** This walker type incorporates wheels on the front two posts to facilitate a step-through gait pattern and faster gait speeds. Glides can be placed on the rear uprights so the user no longer must pick up the walker to advance the device forward.
- **Platform:** This walker type typically involves the addition of forearm troughs to a front-wheeled walker base to facilitate weight bearing through the forearms instead of the hands.
- **Four wheeled/Rollator:** Wheels are located on all four uprights, which allows for a step-through gait pattern. The front wheels may swivel to facilitate turning and maneuvering in tight spaces. This type of walker typically includes a seat for rest breaks during ambulation and hand brakes for greater security for stopping and during sit-to-stand activities.

Fitting and Use Considerations

Walkers generally provide added support and security compared with canes and crutches, with the tradeoff being a slower gait. This is especially true for walkers that do not have wheels. The proper height of the walker is determined by having the patient stand upright with the shoulders relaxed and the elbows flexed 20 degrees.^{23,24} The front of the walker should be maintained approximately 12 inches in front of the patient during standing and ambulation activities.

Pediatric Orthoses

Although pediatric lower extremity orthoses look and function similar to their adult counterparts, they are in reality completely different. A complex set of pathologies coupled with the characteristics of physical growth in a child present additional challenge. A full multidisciplinary team is needed to address the complexities with this patient group. Essential to the evaluation and treatment planning are parents who see the day-to-day changes in their children and who are willing to bring them to the many appointments needed for physician, therapy, and orthotic interventions. Pediatric orthoses are separated into two categories with overlapping groupings of diagnoses: orthopedic problems and neuromuscular problems.

Orthopedic problems can range from fracture management to bilateral hip dysplasia (Fig. 13.29) and congenital talipes



• **Fig. 13.29** SWASH Allard bilateral hip orthosis. (Courtesy CAMP Scandinavia AB.)



• **Fig. 13.30** Dennis Brown bar for club foot. (From Fox JR, Lovegreen W. Lower limb orthoses. In: Webster JB, Murphy DP, eds. *Atlas of Orthoses and Assistive Devices*. 5th ed. Elsevier; 2019:239-246.)

equinovarus (club foot) (Fig. 13.30). Interventions can include surgery, orthoses, and therapy. These are used in combination or separately as needed.²¹

Cerebral Palsy

International experts have defined cerebral palsy (CP) as being a group of permanent but not unchanging disorders of movement, posture, and motor function, which are caused by a nonprogressive interference, lesion, or abnormality of the developing/immature brain. It is not an etiological diagnosis but a clinically descriptive term. It is the most common cause of severe physical disability in childhood, with an overall incidence between 1.5 and 3 children per 1000.

Spastic diplegia is the most common form of CP and accounts for 35% of those affected. It is caused by damage to immature oligodendroglia, and both motor corticospinal and thalamocortical pathways are affected. Fortunately, those children with spastic diplegia have normal cognitive function and good prognosis for independent ambulation. In cases with more severe periventricular leukomalacia, it is possible for spastic quadriplegia to occur. These children often have significant functional limitations, cognitive deficit, epilepsy, visual impairment, and are unlikely to obtain independent ambulation. Spastic hemiplegia is a subtype of CP often seen in infants born at term gestations with a history of in utero or perinatal stroke who usually go on to have normal cognition, high-level functional abilities, and independent ambulation.^{2,54,70,73}

Selective motor control enables agile and independent control of joint motion. Children with CP have impaired selective motor control, which typically occurs in combination with muscle weakness, spasticity, and short muscle-tendon length. This leads in involuntarily coupled movement or joint torques.¹³

Early diagnosis of CP is essential so that interventions can be initiated to optimize their impact on the developing brain's neuroplasticity. Multiple classification systems have been developed to standardize ways to assess support and therapy needs. Therapy is the mainstay of treatment. Physical therapy exercises are designed to improve balance, postural control, and gait and to assist with mobility and transfers. From an occupational therapy standpoint, the focus should be improving fine motor function of the upper extremities to assist the patient in performing activities of daily living (ADLs).^{2,54}

Contracture is a common complication in children with spastic CP, and physical therapy is required for prevention. Serial casting may be useful in reducing or eliminating early moderate contractures. Another mainstay of treatment is botulinum toxin injection to help with focal spasticity. In addition to joint contractures, other common orthopedic problems in CP include hip dysplasia, scoliosis, rotational deformities (including femoral anteversion and tibial torsion), patella maltracking, and foot deformities (including equinovarus and equinovalgus). Required orthoses may include shoe inserts (Fig. 13.31A), AFOs (see Fig. 13.31B), KAFOs (see Fig. 13.31C), HKAFOs, lumbar, thoracolumbosacral, and hand splints. For severe cases of CP and contractures, orthopedic surgeries or dorsal rhizotomy may be required.

Charcot-Marie-Tooth Disease

Charcot-Marie-Tooth (CMT) disease is a group of inherited heterogeneous motor and sensory neuropathy disorders that cause damage to peripheral nerves in the arms and legs either directly or indirectly through myelin sheath destruction. It occurs in 1 in 2500 people and is characterized by distal muscle wasting and weakness, sensory disturbance, hyporeflexia, and pes cavus foot deformity. Symptoms usually begin in the feet and legs during adolescence or early adulthood. CMT disease is usually slowly progressive, but it does not shorten the lifespan of affected individuals. There is weakness in the legs, ankles, and feet with accompanying loss of muscle bulk. Other signs include high foot arches, hammertoes (curled toes), decreased ability to run, foot drop, awkward or higher than normal gait, frequent tripping and falling, and decreased sensation or loss of feeling in the legs and feet.^{43,51}

CMT disease is best managed with therapies and orthoses to assist in mobility and maintain neutral alignment and ROM. As the disease progresses, AFOs are commonly used to improve toe



• **Fig. 13.31** Cascade DAFO foot (A), ankle-foot (B), and knee-ankle-foot orthoses (C). (Courtesy Cascade Dafo, Inc., Ferndale, WA.)

clearance during gait as well as position and ROM of the ankle. In addition, wrist-hand orthoses (WHOs) may also be necessary to manage posture and functional position of the wrists and hands. Orthotic management is used in combination with physical and occupational therapies to maintain available strength and ROM, while working to progress toward safe and independent mobility.³⁷ In recent years, research has become focused on possible gene therapy treatments for CMT disease, including genetic silencing, gene replacing, and correction of metabolic abnormalities.^{74,86}

Arthrogryposis

Arthrogryposis is a nonprogressive neuromuscular disorder often discovered in utero or at birth. It is a term that describes a variety of conditions involving multiple joint contractures or stiffness. The cause is unknown, and the severity can vary. The involved joints often have little to no degrees of ROM. Arthrogryposis occurs in 1 in 3000 live births in the United States, and there is no known cause. Once an adequate diagnosis is made, therapy and orthotic management are necessary to improve function and mobility. Common presentations are flexed upper extremities combined with extended lower extremities or extended upper extremities combined with flexed lower extremities. Distal joints are usually more involved than proximal. The most important early-on (0–3 years of age) treatment goals are to achieve and

maintain optimal ROM. This is achieved through serial casting and the use of orthoses, paired with functional strengthening and a positioning program. It is necessary to take great care to achieve appropriate stretching and splinting because progressing through a program aggressively can cause joint capsule damage or result in a fracture. Rigid splinting is generally avoided in the lower extremities unless it is used to facilitate gait.¹⁰ As children age with arthrogryposis, they often develop secondary problems related to overuse and then may require functional splinting to maintain and preserve ROM.

The most common orthotic management for arthrogryposis for upper extremities are WHOs to improve neutral alignment for improved function, elbow immobilizers to achieve improved extension for functional reach, ADLs, and support during swing phase in gait. Lower extremity orthoses vary with severity but can be KAFOs, AFOs, or knee immobilizers, either in isolation or combination. Additionally, a TLSO may be needed to maintain trunk and spinal alignment to improve posture, breath support, and gastrointestinal motility. In the most severe cases, surgical correction may be needed.

Positional Contractures

Positional contractures refer to a group of conditions in which there is a shortening and hardening of the muscles, tendons,

or other soft tissues around a joint, leading to rigidity and loss of motion of the joint.⁹¹ Contractures may occur for several reasons but are common in patients with increased muscle tone or spasticity and/or decreased mobility, including those who are primary wheelchair users. For example, patients with increased muscle tone or spasticity often present with posturing in flexed positions. Upper extremities with increased tone often result in elbow, wrist, and finger flexion combined with shoulder internal rotation. Lower extremities commonly rest in hip and knee flexion combined with ankle plantarflexion. Without intervention, such as passive stretching or positioning with orthoses, these joints can become rigid in a flexed position and lack full extension ROM. This is often common in patients who have decreased mobility or those who sit or use a wheelchair as their primary mode of mobility. Sitting for extended periods of time can lead to shortening of the hip flexor muscles, knee flexors, and plantar flexors, resulting in decreased ROM or contractures in hip flexion, knee flexion, and ankle plantarflexion. Orthoses such as AFOs and knee immobilizers can be used to stretch and improve ROM or to position and decrease the risk of further contracture. Similar orthoses of the upper extremities, such as WHOs and elbow immobilizers, may also be used for dynamic mobility or as resting splints to maintain more neutral alignment.⁹¹ Orthoses can be made to offer dynamic joint stretching. This is often done with nighttime AFOs for the ankle as well as dynamic knee immobilizers for the knees. Similar upper extremity dynamic orthoses exist to stretch the elbow. In cases of severe positional contractures, serial casting or surgery may also be appropriate.

Idiopathic Club Foot

Congenital talipes equinovarus, or idiopathic club foot, affects nearly 1 to 2 infants per 1000 live births and is the most common serious musculoskeletal birth defect in the world. It is seen in higher rates in males and can be bilateral in up to 50% of those affected. Idiopathic club foot is a separate entity from the benign positional leg and foot anomalies that are commonly caused by third-trimester crowding, resolve on their own, and include metatarsus adductus, calcaneovalgus, and positional cavovarus.^{52,91}

The primary cause of idiopathic club foot is Achilles tendon shortening. Four key components of the deformity exist: a high arch, the forefoot being turned in, the heel being turned in, and the hind foot being pointed downward. These deformities cannot be corrected with gentle manipulation, and the affected foot and calf may be slightly smaller than the nonaffected side. The mainstay of treatment is the Ponseti method, which involves two stages. The first stage involves casting, and the goal is to correct the foot deformity. The first cast is placed 1 to 2 weeks after birth, and casts must be changed every 7 to 10 days for a total of four or five casts. Before placement of the last cast, a tenotomy of the Achilles tendon is performed. The last cast is worn for 2 to 3 weeks. The second, or bracing, stage is initiated at this point. Ponseti bracing is aimed at maintaining the correction obtained in the casting stage. The baby is fit in a supramalleolar orthosis with a bar (see Fig. 13.30). This orthosis is worn for 23 hours a day for the first 2 months and then 12 hours a day until kindergarten age. If the Ponseti method is strictly adhered to, there is a 90% to 100% initial correction rate.^{12,52,91} Although recurrence is rare, it usually begins with a tightening of the heel cord. This leads to an in-toeing gait with increased adduction of the forefoot, varus of the heel, and loss of heel strike.⁵²

Genu Recurvatum

Genu recurvatum refers to a condition in which the tibiofemoral joint extends beyond neutral or into knee hyperextension greater than 5 degrees.¹⁹ Children presenting with genu recurvatum may exhibit knee pain, have an extension gait pattern, and have poor control during the terminal stance phase of gait. The posterior structures of the knee are also often placed under increased stress with this condition. Treatment approaches are dependent on the severity of the hyperextension but consist of strengthening of the quadriceps muscles to improve control, using a splint or brace to block knee extension past neutral and maintain slight flexion, or using an AFO to improve control. AFOs are the most common orthosis used with genu recurvatum gait because the long lever arm works to maintain control over the knee when the stance phase begins, lessening the hyperextension moment.³⁷

Congenital Hip Dysplasia

Hip dysplasia occurs when the hip joint does not develop properly, and the acetabulum is too shallow. This allows the femoral head to slip partially or completely out of joint. Dysplasia can range from mild abnormalities to complete dislocation of the hip. If left untreated, hip dysplasia can lead to irreversible pain and loss of function later in life caused by increased loads being applied to an abnormally small surface area. Outside of the newborn stage, other indicators of hip dysplasia include leg length discrepancy, uneven skin folds on the thighs, and decreased mobility and flexion on one side.

The goal of hip dysplasia treatment is to obtain and maintain a concentric reduction of the femoral head into the acetabulum to allow for continued normal development. Different ages of patients require different kinds of treatment. Infants up to 6 months of age with confirmed instability or dislocation are initially treated using a Pavlik harness (Fig. 13.32), which is used to hold the hips in a position of flexion and abduction. This allows for centering of the femoral head in the acetabulum.^{7,66,88}

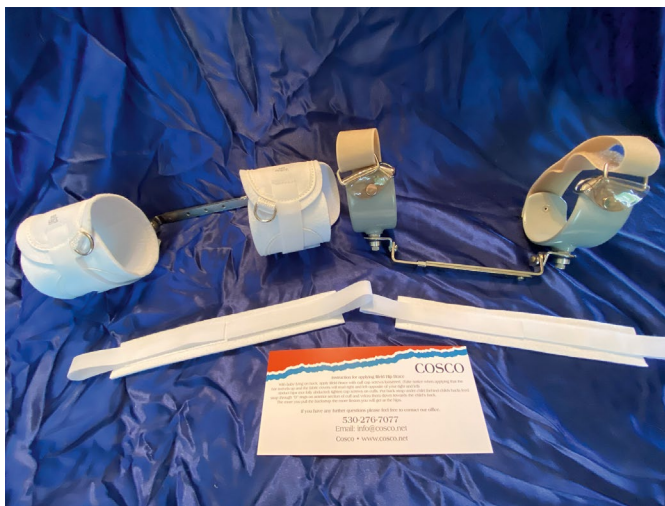
If the Pavlik harness fails to correct the dysplasia, many times infants are placed in an Ilfeld orthosis (Fig. 13.33), which allows the hips to be held more rigidly in abduction. Older infants must often undergo closed reduction with the hip placed in 90 to 100 degrees of flexion and the minimal amount of abduction needed to maintain a stable hip reduction. They are then placed in a spica cast for healing. After 12 months of age, into adolescence, children diagnosed with hip dysplasia must undergo open surgical reduction and repair to ensure ongoing normal gait and function and to avoid arthritis later in life.^{7,66}

Muscular Dystrophy

Muscular dystrophy is a broad term that refers to a group of disorders caused by multiple different genes, which leads to damage of the integrity of muscle fibers throughout the body. This damage can be profound and lead to muscle degeneration, muscle fiber death and phagocytosis, fiber branching and splitting, and eventual replacement of muscle tissue with fibrous tissue and fat. These changes lead to progressive weakness and loss of muscle mass. Initial symptoms often progress and lead to further complications, such as trouble walking without assistance, difficulty using one's arms, joint contractures, difficulty breathing, scoliosis, heart problems caused by inefficiency of cardiac muscle, and swallowing problems.^{16,53,67}



• **Fig. 13.32** Pavlik harness.(Courtesy Fillauer, LLC, Chattanooga, TN.)



• **Fig. 13.33** Dynamic Ilfeld hip abduction splint. (Courtesy COSCO, Christensen Orthopedic Supply Company, Redding, CA.)

Treatment is largely focused on medications and therapy. Therapy is focused on ROM, stretching, and low-impact aerobic exercise. Bracing can be used to keep muscles and tendons stretched and flexible to slow progression of contractures. Braces are also used to aid in mobility and function by supporting weakened muscles. Despite the previously mentioned treatments, almost all patients will progress to a stage in which they depend on mobility aids and will require assistance with breathing and oxygen support.

The Cascade DAFO Company (see Fig. 13.31), which has made pediatric orthoses their primary area of focus, provides comprehensive orthotic selection reference guides.⁴¹ In general, treatment of children requires a team of physicians, therapists, and orthotists who are clinically experienced with children and their unique challenges.

Other mobility aids are standing platforms (Fig. 13.34) and wheeled-type walkers (Fig. 13.35). With these, you will see lower, upper, and spinal orthoses used in conjunction with the device. This allows the patient the freedom of mobility and the necessary functional support.



• **Fig. 13.34** Hugs Stander vertical standing platform. (Courtesy Sammons Preston, Bolingbrook, IL.)



• **Fig. 13.35** Trekker Gait Trainer.(Courtesy Drive, Port Washington, NY.)

Summary

Lower limb orthoses play an integral role in the treatment of many conditions of the lower extremity of a neurologic, orthopedic, and congenital nature. Allowing both Adult and Pediatric patients with lower extremity pathologies to interface with the ground and the world around them by improving gait, reducing pain, protection of joints, and control of aberrant movements or spasticity. People with the same diagnoses all present differently and require unique solutions. This is especially true with the pediatric population, as they are not to be considered small adults. The complexities of providing lower extremity orthoses cannot be understated due to the various biomechanical considerations and require a team with holistic view of the goal for that patient.

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